hood is drawn around its parent's with weight ρ and around the mean with weight $1-\rho$, so ρ is how much of its parent's current productivity a child keeps. It is the channel through which anything that lowers a parent's productivity lowers its child's: a parent whose own productivity carries the penalty δd, because its own parent had a SUD, also passes a lower draw on to its child, so a parent's SUD lowers productivity beyond the next generation. Raise it and a child's prospects depend more on its parent's decisions, and the earnings externality grows. It is set externally at 0.50, between the intergenerational elasticities of earnings in Solon (1992) and Mazumder (2005). $\bar{\pi}$eu, the job separation rate. This parameter determines the chance that an employed household loses

its job each year. It is scaled by ϖ(y) in (21), so separations are likelier at low productivity. Raise it and more households lose work each year, so the unemployment rate rises with it. It is chosen to reproduce the aggregate unemployment rate in the Current Population Survey (CPS) of the Bureau of Labor Statistics (BLS) over 2014 to 2024, 4.9 percent, given the job-finding rate set externally below. ϖ(y), separations by productivity. This parameter tilts the separation rate across productivity, so that

low-productivity households lose their jobs more often, which is what puts them at a higher risk of homelessness. Steepen the tilt and job loss concentrates further among the poorest, and so does the risk of homelessness. It is normalised to average one across households, so that $\bar{\pi}$eu alone sets the aggregate unemployment rate. Each productivity node takes the unemployment odds of the education group it falls in, from the BLS 2019 annual averages of unemployment by educational attainment for ages 25 and over, with the education groups ordered along productivity by their median earnings. $\bar{\pi}$ue, the annual job-finding rate. This parameter is the probability that an unemployed household finds

work each year it is unemployed. With the divisors κue

d and κue

m it sets how long an unemployed household with a SUD or SMI is expected to remain unemployed. Raise it and spells out of work are shorter on average, so fewer households fall behind on the rent. I set it to 0.55 from the annual exit rate Krueger et al. (2014) estimate on matched CPS panels. pui, the share of separations that receive UI. A household that loses its job draws UI for one year

with probability pui and nothing otherwise. It sets how many of the newly unemployed fall straight to zero earnings, and how costly being unemployed is. I set it to 0.30, since in the BLS supplement to the CPS for 2022 about seven in ten unemployed people who worked in the past year did not apply for UI after losing their job.

Block 10: Baseline survival and SMI onset. These two parameters are the survival and the SMI-onset age profiles.

$\bar{\phi}$(j), baseline survival by age. This is the baseline chance of surviving to the next age, before any mul-

tiplier. It is the base that every mortality multiplier in Section 5, for a SUD, homelessness

and SMI, scales. Raise it and households expect to live longer, so the life a SUD puts at risk is worth more. I take the profile from the life table for the total population of the United States in 2010 (Arias, 2014), for ages 26 to 85. ϛm(j), the age profile of SMI onset. This determines the shape of onset across ages, normalised to one

at age 26, so that ωm in Table 3 sets its level alone. It is here rather than in Table 3 because, like the age survival profile, it only sets the baseline. What actually affects the impact of Housing First versus Treatment First is ι, the multiplier a SUD puts on SMI onset, which is in Table 3. It is the onset that reproduces a log-linear fit to the share of adults carrying SMI at each age in NSDUH 2024.

Block 11: Interest and taxes.

r, the risk-free rate. This parameter sets the return on saving. Raise it and saving pays more, so house-

holds hold more assets to carry them through a spell out of work or without a voucher. I set it to three percent, from Kaplan et al. (2020).

φ, tax progressivity. This is the exponent on income in the tax schedule of Section 4.3, and it determines

the progressivity of the labour income tax. Lower it and the tax is more progressive, so households with low incomes, among them most voucher holders and most drug users, keep more of what they earn. It is set to 0.85, in line with Heathcote et al. (2017). τℓ, the scale of after-tax labour income. Given φ, this parameter sets how much of their labour income

households keep after tax on average, so raising it lowers the tax liability. It does not move in the experiments: the budget is balanced by πV instead. It is calibrated to an average labour income tax rate of 22.4 percent, which is what McDaniel (2007) measures for the United States and what Guvenen et al. (2023) use. τk, the capital income tax rate. This parameter affects the after-tax return on savings. Raise it and

saving pays less, so households hold fewer assets to fall back on. I set it to fifteen percent, consistent with the statutory United States rate on long-term capital gains and qualified dividends for the middle of the income distribution, which the Internal Revenue Service (IRS) sets out in Topic 409 (https: //www.irs.gov/taxtopics/tc409).

Block 12: Social security benefits and programmes’ rules.

S(y) and S, social security and its floor. Retirement income follows the statutory primary insurance

amount: it replaces a share of the household’s income at its last working age, a share that falls at each of the two bend points of the Social Security Administration (SSA), up to the taxable maximum, with the SSI payment as a floor (the full function is detailed in footnote 16). As S(y) rises the old are richer, so the value of a life still to be lived rises and with it the cost of a disorder that shortens it. The shares α1, α2, α3 are 0.90, 0.32, 0.15 below the first bend point, between the two and above the second, respectively. The bend points k1

and k2 and the taxable maximum ymax are the SSA's 2024 dollar amounts divided by its average wage index, 0.22, 1.33 and 2.64, respectively. The floor, S, is 0.18 of average labour income. ψui and ¯y, UI. These two set the UI replacement rate and the cap on the earnings it is paid on, respectively.

Raise either and the first year out of work costs a household less, so a spell out of work is less likely to end in a shelter. I set them to fifty percent and one times average labour income, respectively, taking both from the statutory replacement rates and state benefit maxima that the Department of Labor compiles in its comparison of state UI laws (https:// oui.doleta.gov/unemploy/statelaws.asp). κV and κr, the voucher rules. These two set the income cutoff for eligibility and the tenant's rent contri-

bution, respectively. Raise κV and more households are eligible, so more have a voucher to lose under Treatment First; raise κr and a voucher pays less, so losing one costs less. I take both from HUD's own rules: half of median income and thirty percent, respectively. The first is the very low-income limit HUD publishes annually with its income limits (https:// www.huduser.gov/portal/datasets/il.html); the second is the share of its before-tax income a tenant pays towards the rent, the voucher pay- ing the rest up to the payment standard, under the Housing Choice Voucher programme (https: //www.hud.gov/helping-americans/housing-choice-vouchers). ϱ, the food stamp clawback rate. This is the rate at which food assistance is withdrawn as income rises.

Raise it and food assistance is clawed back faster, so it stops at a lower income and pays less to every household that still receives it. I set it to thirty percent, which is the statutory rule (7 CFR 273.10).

Block 13: Drug market. The two parameters in this block describe the rest of the drug market of Block 6. The first, JE, fixes who can enter the market (footnote 21); the second is set so that supply equals demand in the benchmark, and is then held fixed.

JE, the last age at which use can begin. This parameter is the last age at which a household without

a SUD can start using; one that already has a SUD at that age can keep using. It decides who can enter the drug market, and so how many households a policy that deters use can reach. As it rises, more old households can start using at little cost, since too little life remains for the disorder to damage (footnote 21), and the model produces too much old-age use. I set it to model age 55, the equivalent of age 80 when the model runs to age 85.

ν1, the supply shifter. A pure normalisation, chosen so that supply equals demand at pv = 1 in the

benchmark, and kept fixed across experiments, in which the price is free to adjust. It normalises units, not behaviour.